

4-10 The spinning disk

In the last lecture (see *FLP* Vol. I, Ch. 20, “Rotation in Space”) we discussed the interesting fact that the angular momentum of a rigid body is not necessarily in the same direction as its angular velocity. We took as an example a disk that is fastened onto a rotating shaft in a lopsided fashion, as shown in Figure 4-26. I’d like to explore this example in further detail.

First, let me remind you of an interesting thing that we’ve already talked about: that for any rigid body, there is an axis through the body’s center of mass about which the moment of inertia is maximal, there is another axis through the body’s center of mass about which the moment of inertia is minimal, and these are always at right angles. It’s easy enough to see this for a rectangular block as shown in Figure 4-27, but surprisingly it’s true for any rigid body.

These two axes, and the axis which is perpendicular to them both, are called the principal axes of the body. The principal axes of a body have

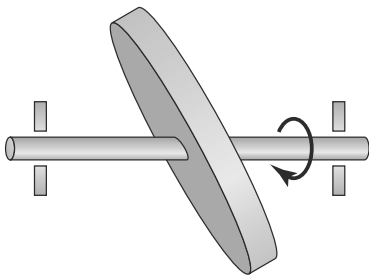


FIGURE 4-26 A disk fastened in a lopsided manner to a spinning shaft.